

1. Stored procedure

```
create procedure spGetNamesGender
as begin
    select FirstName, LastName, Gender from dbo.DimEmployee
end

execute spGetNamesGender
```

Sellel ma saan kohe pärida töölise esinime, perekonnanime ja sugu

Results		Messages	
	FirstName	LastName	Gender
1	Guy	Gilbert	M
2	Kevin	Brown	M
3	Roberto	Tamburello	M
4	Rob	Walters	M
5	Rob	Walters	M
6	Thierry	D'Hers	M
7	David	Bradley	M
8	David	Bradley	M
9	JoLynn	Dobney	F
10	Ruth	Ellerbrock	F
11	Gail	Erickson	F
12	Barry	Johnson	M
13	Jossef	Goldberg	M
14	Terri	Duffy	F
15	Sidney	Higa	M
16	Taylor	Maxwell	M
17	Jeffrey	Ford	M
18	Jo	Brown	F
19	Doris	Hartwig	F
20	John	Campbell	M
21	Diane	Glimp	F
22	Steven	Selikoff	M
23	Peter	Krebs	M
24	Stuart	Munson	M
25	Greg	Alderson	M
26	David	Johnson	M
27	Zheng	Mu	M
28	Ivo	Salmre	M
29	Paul	Kemczinski	M

✓ Query executed successfully.

2. Stored procedure eeliseks on see kui on päring mida võibolla on vaja tihti kasutada, siis oleks lihtsam luua stored procedure, et teha töö kiiremaks

3. Funktsioon

```
create function dbo.CalculateAge (@DOB date)
returns int
as begin
declare @Age int

set @Age = datediff(year, @DOB, getdate()) -
    case
        when (month(@DOB) > month(getdate())) or
             (month(@DOB) = month(getdate()) and day(@DOB) > day(getdate()))
        then 1
        else 0
        end
    return @Age
end
```

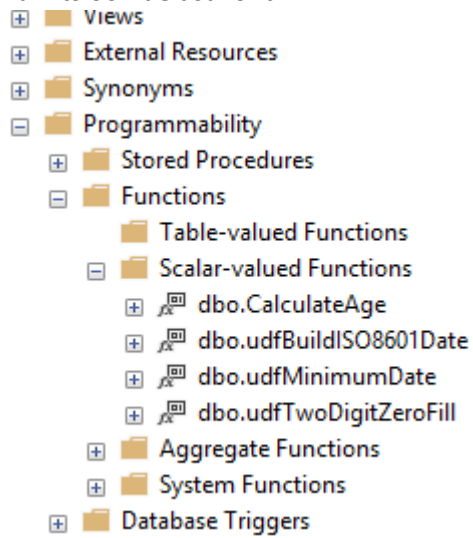
```
select employeekey, FirstName, Lastname, dbo.CalculateAge(BirthDate) as Age from
dbo.DimEmployee
```

Arvutab isiku vanuse

Results		Messages		
	employeekey	FirstName	Lastname	Age
1	1	Guy	Gilbert	44
2	2	Kevin	Brown	39
3	3	Roberto	Tamburello	51
4	4	Rob	Walters	51
5	5	Rob	Walters	51
6	6	Thierry	D'Hers	67
7	7	David	Bradley	51
8	8	David	Bradley	51
9	9	JoLynn	Dobney	70
10	10	Ruth	Ellerbrock	70
11	11	Gail	Erickson	74
12	12	Barry	Johnson	70
13	13	Josief	Goldberg	67
14	14	Teri	Duffy	55
15	15	Sidney	Higa	70
16	16	Taylor	Maxwell	70
17	17	Jeffrey	Ford	70
18	18	Jo	Brown	70
19	19	Doris	Hartwig	70
20	20	John	Campbell	70
21	21	Diane	Glimp	70
22	22	Steven	Selikoff	49
23	23	Peter	Krebs	43
24	24	Stuart	Munson	64
25	25	Greg	Alderson	50
26	26	David	Johnson	46
27	27	Zheng	Mu	42
28	28	Ivo	Salmre	44
29	29	Paul	Kempinski	45

✓ Query executed successfully.

4. Funktsioonide asukoht



5. Funktsiooni on võimalik krüpteerida

6. Local temp tabel

```
create table #PersonHobbies(Id int, Name nvarchar(20))
insert into #PersonHobbies values(1, 'Walking')
insert into #PersonHobbies values(2, 'Running')
insert into #PersonHobbies values(3, 'Swimming')
go

select * from #PersonHobbies
```

Results		Messages
Id	Name	
1	Walking	
2	Running	
3	Swimming	

Query executed successfully.

7. Global temp tabel

```
create table ##GlobalPersonHobbies(Id int, Name nvarchar(20))
insert into ##GlobalPersonHobbies values(4, 'Sprinting')
insert into ##GlobalPersonHobbies values(5, 'Gaming')
insert into ##GlobalPersonHobbies values(6, 'Singing')
go

select * from ##GlobalPersonHobbies
```



	Id	Name
1	4	Sprinting
2	5	Gaming
3	6	Singing

8. Global ja local temp tabeli vahe on see, et globaalset tabelit saab kasutada ja näha kõigis sessioonides, aga lokaalne on ainult selles sessioonis kus see loodi.
9. Indeksi luuaks ja kasutatakse, et teha töö ja päringute tegemine kiiremaks

10.

```
create index IX_Customer_YearlyIncome
on dbo.DimCustomer(YearlyIncome desc)

select FirstName, LastName, YearlyIncome from dbo.DimCustomer with (index
(IX_Customer_yearlyIncome))
```

See indeks paneb kliendi aasta sissetuleku kahanevasse järjekorda

	FirstName	LastName	YearlyIncome
2...	Erik	Gomez	100000,00
2...	Devin	Peterson	100000,00
2...	Stephanie	Ross	100000,00
2...	Jocelyn	Gonzales	100000,00
2...	Alexis	Wilson	100000,00
2...	Brandon	Moore	100000,00
2...	Blake	Thomas	100000,00
2...	Anthony	Williams	100000,00
2...	Carlos	Allen	100000,00
2...	Antonio	Henderson	100000,00
2...	Alex	King	100000,00
2...	Timothy	Lopez	100000,00
2...	Willie	Wang	100000,00
2...	Beth	Alvarez	100000,00
2...	Edgar	Perez	100000,00
2...	Jon	Yang	90000,00
2...	Trevor	Bryant	90000,00
2...	Dalton	Perez	90000,00
2...	Cheryl	Diaz	90000,00
2...	Bianca	Lin	90000,00
2...	Bryce	Richardson	90000,00
2...	Carol	Howard	90000,00
2...	Mason	Roberts	90000,00
2...	Nicholas	Thompson	90000,00
2...	Trinity	Richardson	90000,00
2...	Eduardo	Martin	90000,00
2...	Elizabeth	Jones	90000,00
2...	Taylor	Howard	90000,00
2...	Nathan	Lee	90000,00

 Query executed successfully.

11.

```
create view vCustomerByCity
as
select FirstName, LastName, City
from dbo.DimCustomer
join dbo.DimGeography
on dbo.DimCustomer.GeographyKey = dbo.DimGeography.GeographyKey

select * from vCustomerByCity
```

Results		Messages	
	FirstName	LastName	City
1	Jon	Yang	Rockhampton
2	Eugene	Huang	Seaford
3	Ruben	Tomes	Hobart
4	Christy	Zhu	North Ryde
5	Elizabeth	Johnson	Wollongong
6	Julio	Ruiz	East Brisbane
7	Janet	Alvarez	Matrville
8	Marco	Mehta	Warrnambool
9	Rob	Verhoff	Bendigo
10	Shannon	Carlson	Hervey Bay
11	Jacquelyn	Suarez	East Brisbane
12	Curtis	Lu	East Brisbane
13	Lauren	Walker	Bremerton
14	Ian	Jenkins	Lebanon
15	Sydney	Bennett	Redmond
16	Chloe	Young	Burbank
17	Wyatt	Hill	Imperial Beach
18	Shannon	Wang	Sunbury
19	Clarence	Rai	Bendigo
20	Luke	Lal	Langley
21	Jordan	King	Metchosin
22	Destiny	Wilson	Beaverton
23	Ethan	Zhang	Bellingham
24	Seth	Edwards	Bellflower
25	Russell	Xie	Concord
26	Alejandro	Beck	Hawthorne
27	Harold	Sai	Goulburn
28	Jessie	Zhao	Warrnambool
29	Jill	Jimenez	St. Leonards

 Query executed successfully.

```

create view vOrganizationByFactFinance
as
select OrganizationName, Amount, date
from dbo.DimOrganization
join dbo.FactFinance
on dbo.FactFinance.OrganizationKey = dbo.DimOrganization.OrganizationKey

select * from vOrganizationByFactFinance

```

Results		Messages	
	OrganizationName	Amount	date
1	Northeast Division	22080	2010-12-29 00:00:00.000
2	Northeast Division	20200	2010-12-29 00:00:00.000
3	Northeast Division	2000	2010-12-29 00:00:00.000
4	Northeast Division	2208	2010-12-29 00:00:00.000
5	Northeast Division	1546	2010-12-29 00:00:00.000
6	Northeast Division	1800	2010-12-29 00:00:00.000
7	Northeast Division	380	2010-12-29 00:00:00.000
8	Northeast Division	378	2010-12-29 00:00:00.000
9	Northeast Division	344	2010-12-29 00:00:00.000
10	Northeast Division	380	2010-12-29 00:00:00.000
11	Northeast Division	200	2010-12-29 00:00:00.000
12	Northeast Division	174	2010-12-29 00:00:00.000
13	Northeast Division	132	2010-12-29 00:00:00.000
14	Northeast Division	100	2010-12-29 00:00:00.000
15	Northeast Division	38	2010-12-29 00:00:00.000
16	Northeast Division	54	2010-12-29 00:00:00.000
17	Northeast Division	70	2010-12-29 00:00:00.000
18	Northeast Division	300	2010-12-29 00:00:00.000
19	Northeast Division	250	2010-12-29 00:00:00.000
20	Northeast Division	114	2010-12-29 00:00:00.000
21	Northeast Division	100	2010-12-29 00:00:00.000
22	Northeast Division	280	2010-12-29 00:00:00.000
23	Northeast Division	297	2010-12-29 00:00:00.000
24	Northeast Division	535	2010-12-29 00:00:00.000
25	Northeast Division	480	2010-12-29 00:00:00.000
26	Northeast Division	200	2010-12-29 00:00:00.000
27	Northeast Division	153	2010-12-29 00:00:00.000
28	Northeast Division	586	2010-12-29 00:00:00.000
29	Northeast Division	550	2010-12-29 00:00:00.000

✓ Query executed successfully.